COVER PAGE.SchDoc

BLOCK DIAGRAM.SchDoc

CONNECTORS.SchDoc

CPU - USB, PCIE, ETHERNET.SchDoc

CPU - SD, SAI.SchDoc

Power.SchDoc

CPU - LPDDR4.SchDoc

CPU - POWER.SchDoc

CPU - HDMI, CSI, DSI, LVDS.SchDoc

eMMC, QSPI Flash.SchDoc

Ethernet.SchDoc

LPDDR4.SchDoc

CPU - JTAG, CONTROL.SchDoc

Power Seq.SchDoc

Rev History.SchDoc

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IMX8 - Atlas

Variant: [No Variations]

31/08/2026
V1I1

PRELIMINARY

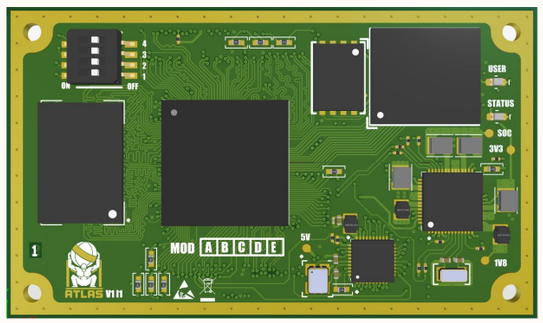
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2	BLOCK DIAGRAM	12	ETHERNET	22	.....	32	.....
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4	CPU - DDR4	14	POWER SEQUENCING	24	.....	34	.....
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6	CPU - USB, PCIE, ETHERNET	16	.....	26	.....	36	.....
7	CPU - POWER	17	.....	27	.....	37	.....
8	CPU - JTAG, CONTROL	18	.....	28	.....	38	.....
9	CPU - HDMI, CSI, DSI, LVDS	19	.....	29	.....	39	.....
10	LPDDR4	20	.....	30	.....	40	.....

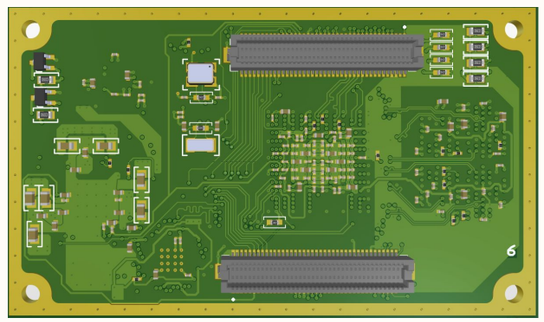
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EXAMPLE TEXT FOR CRITICAL DESIGN NOTES

EXAMPLE TEXT FOR CAUTIONARY DESIGN NOTES

EXAMPLE TEXT FOR CRITICAL LAYOUT GUIDELINES

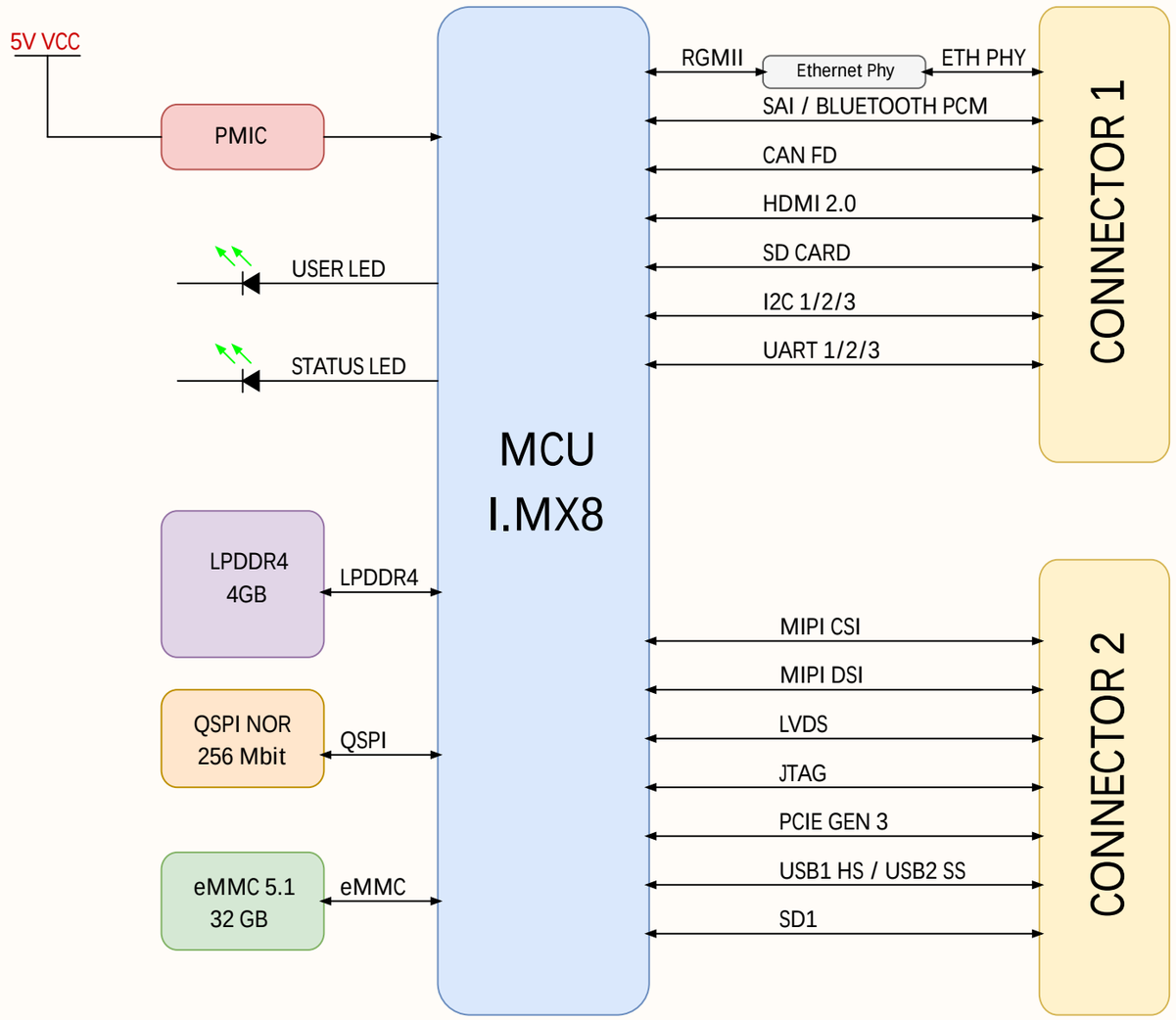




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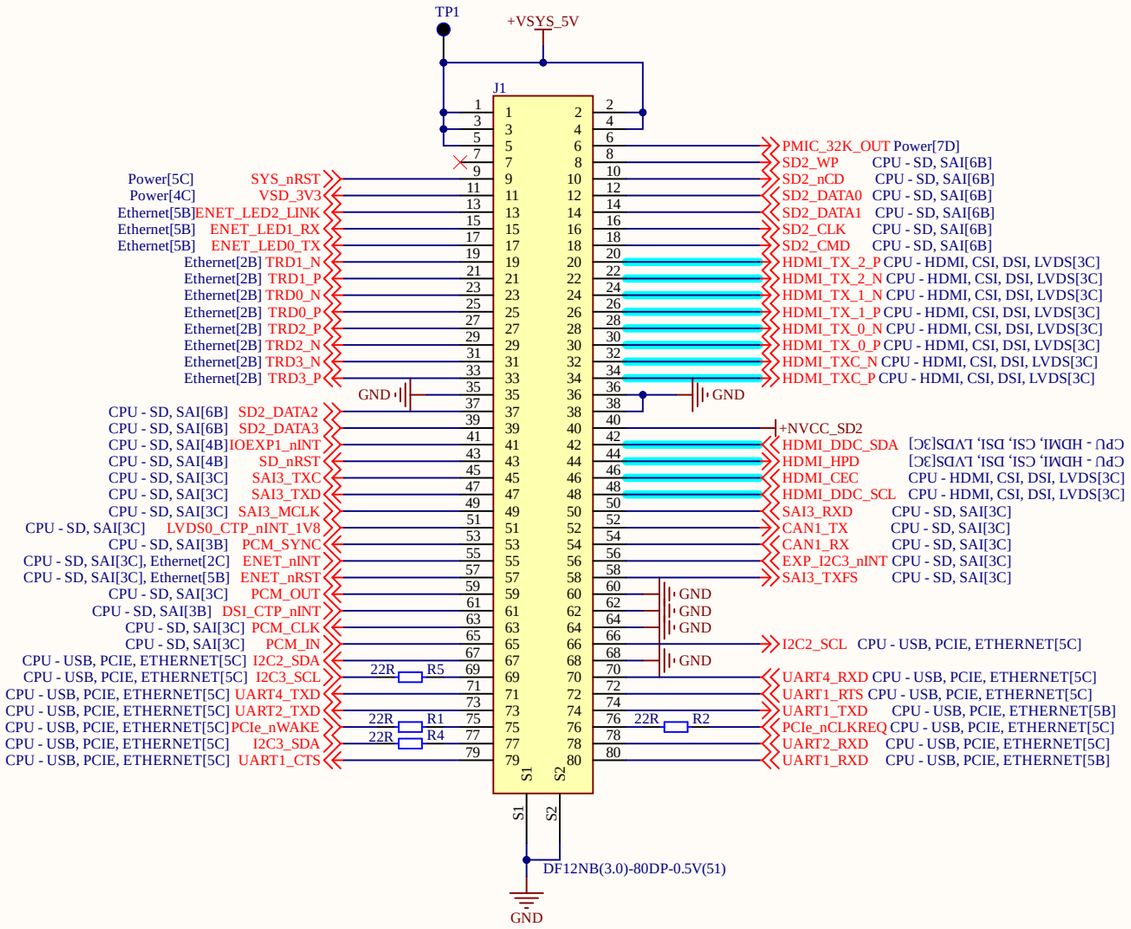
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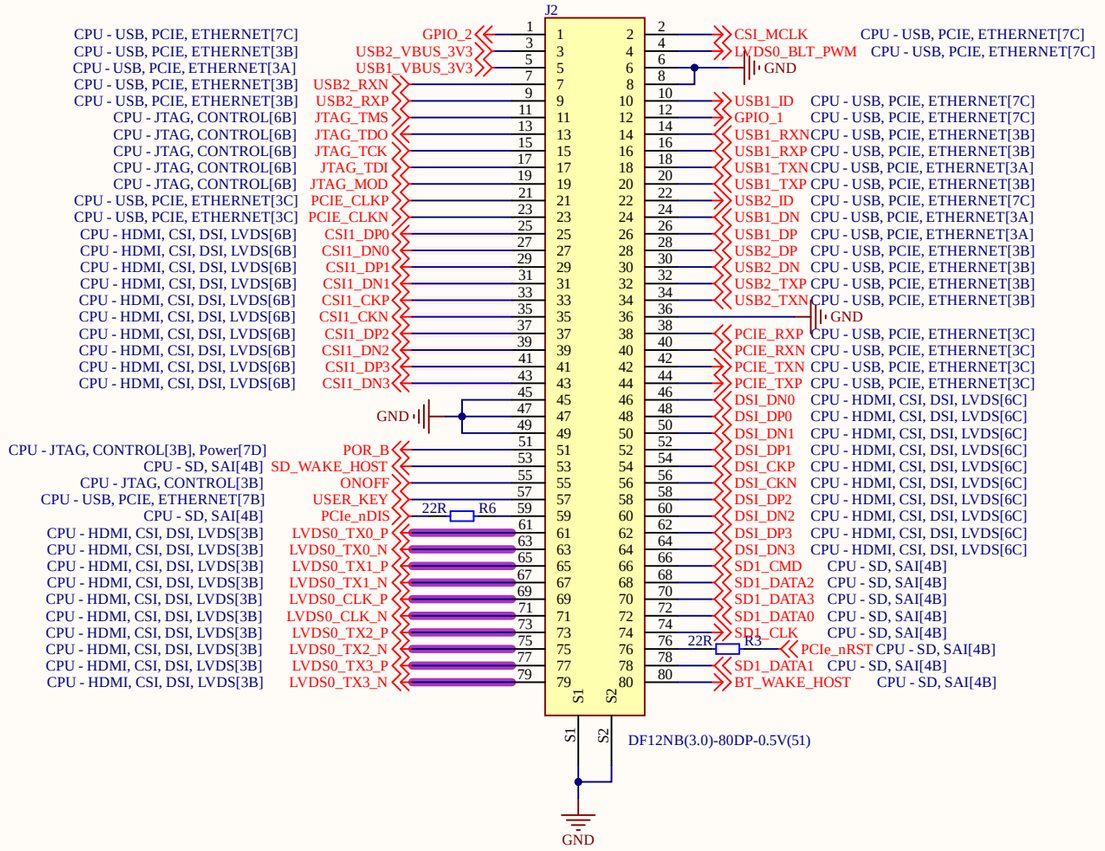
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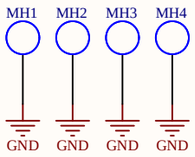


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CONNECTORS









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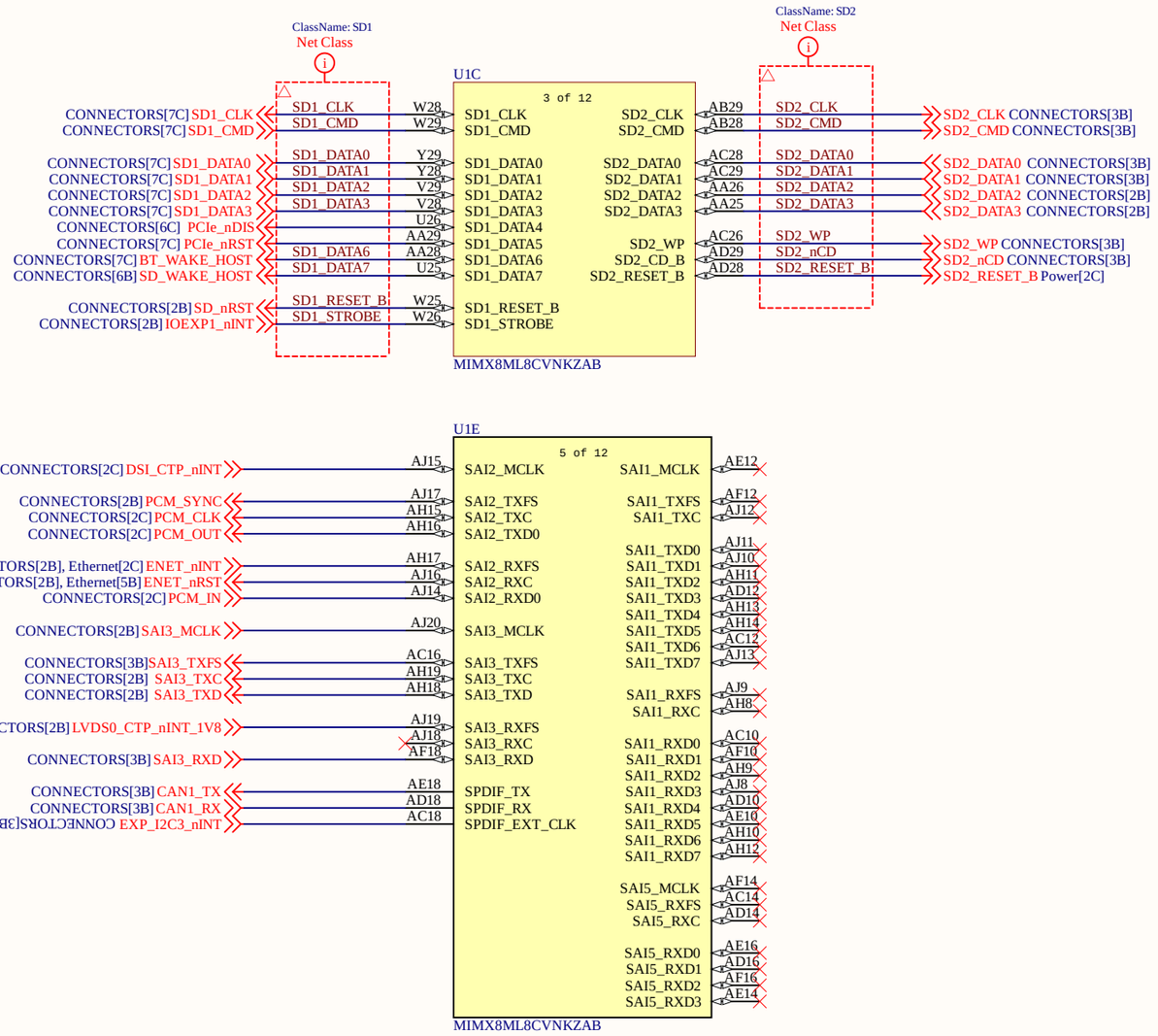
CPU - DDR4



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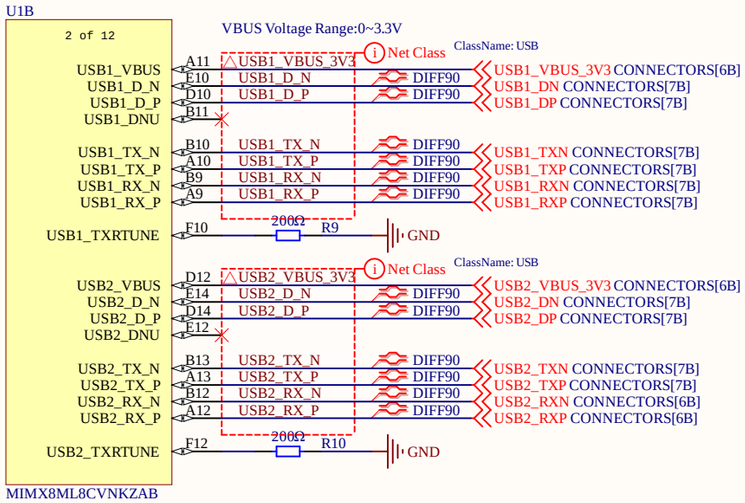
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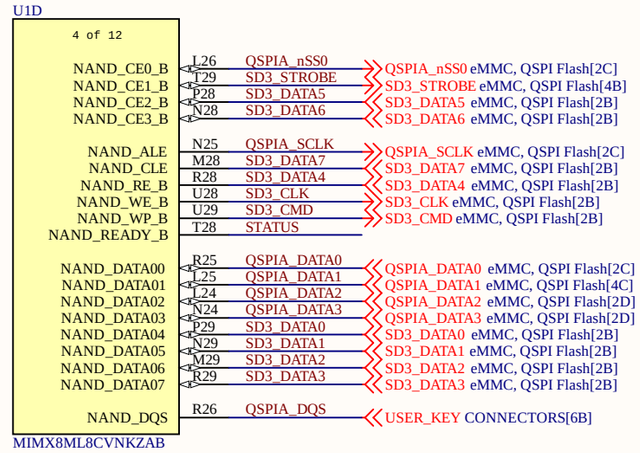
CPU - SD, SAI

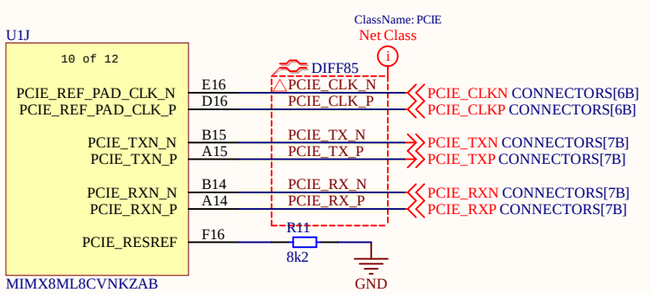


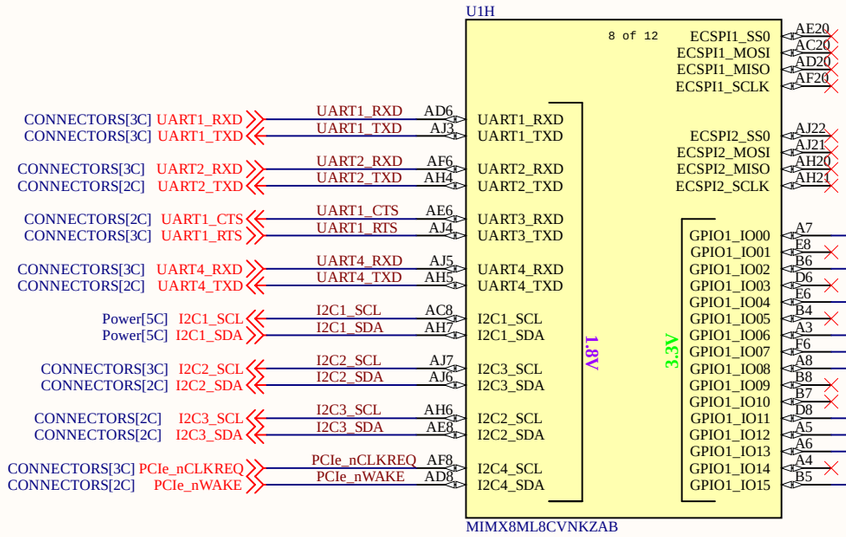
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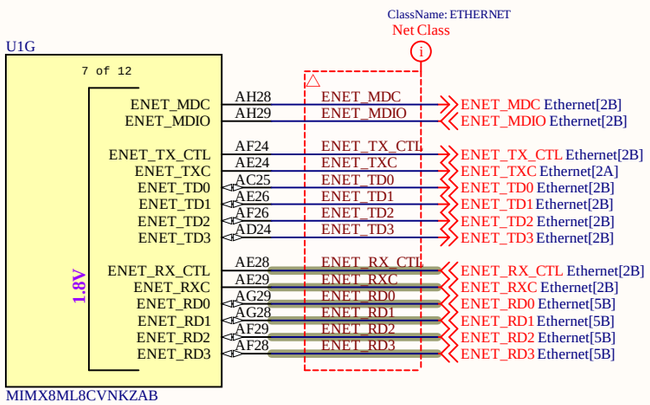
CPU - USB, PCIE, ETHERNET

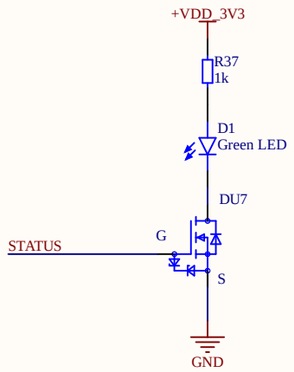


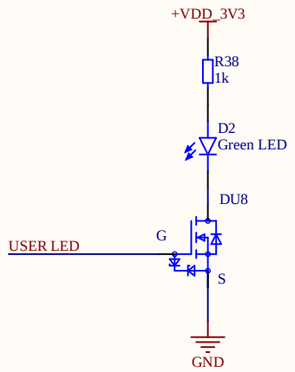








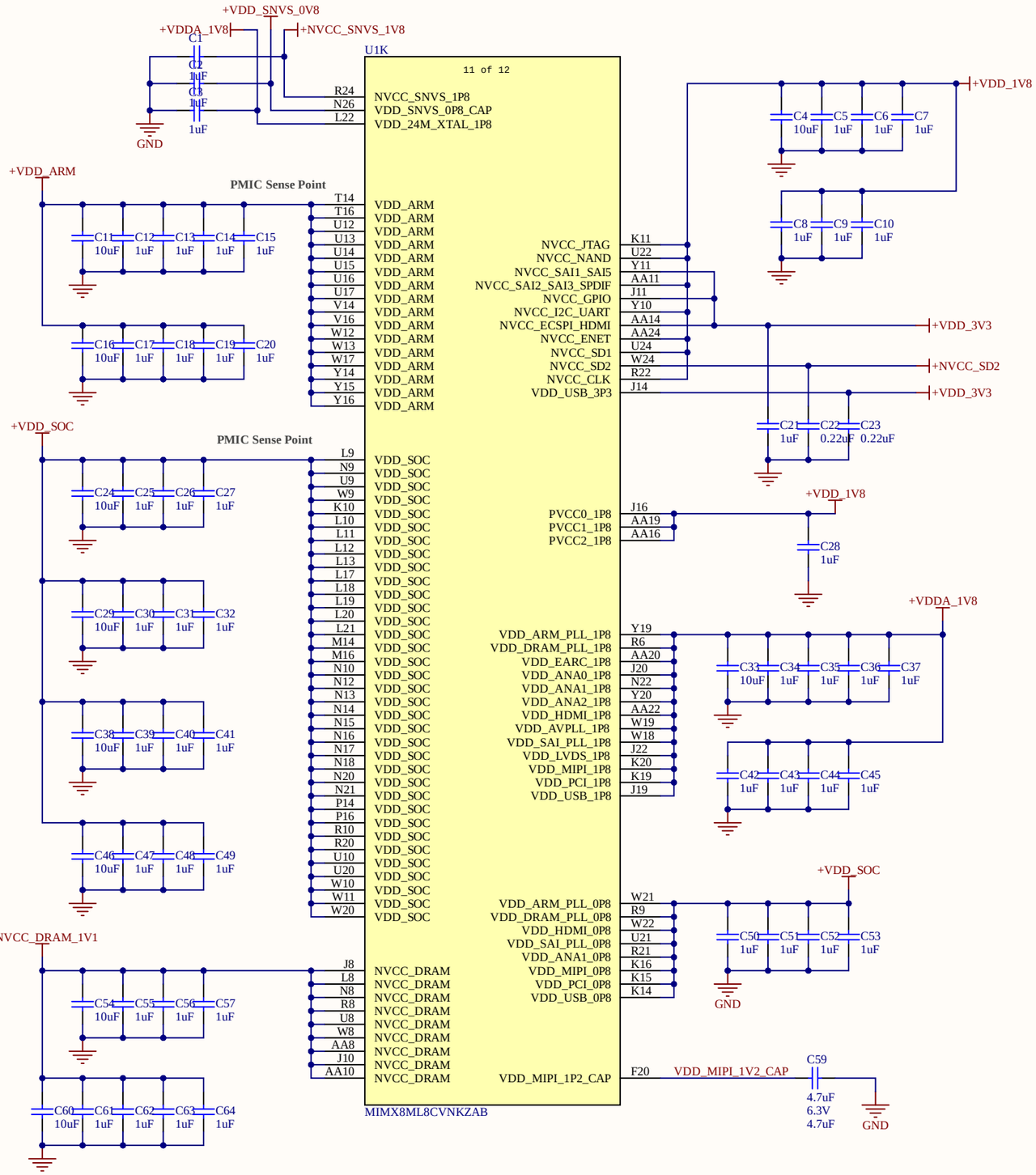




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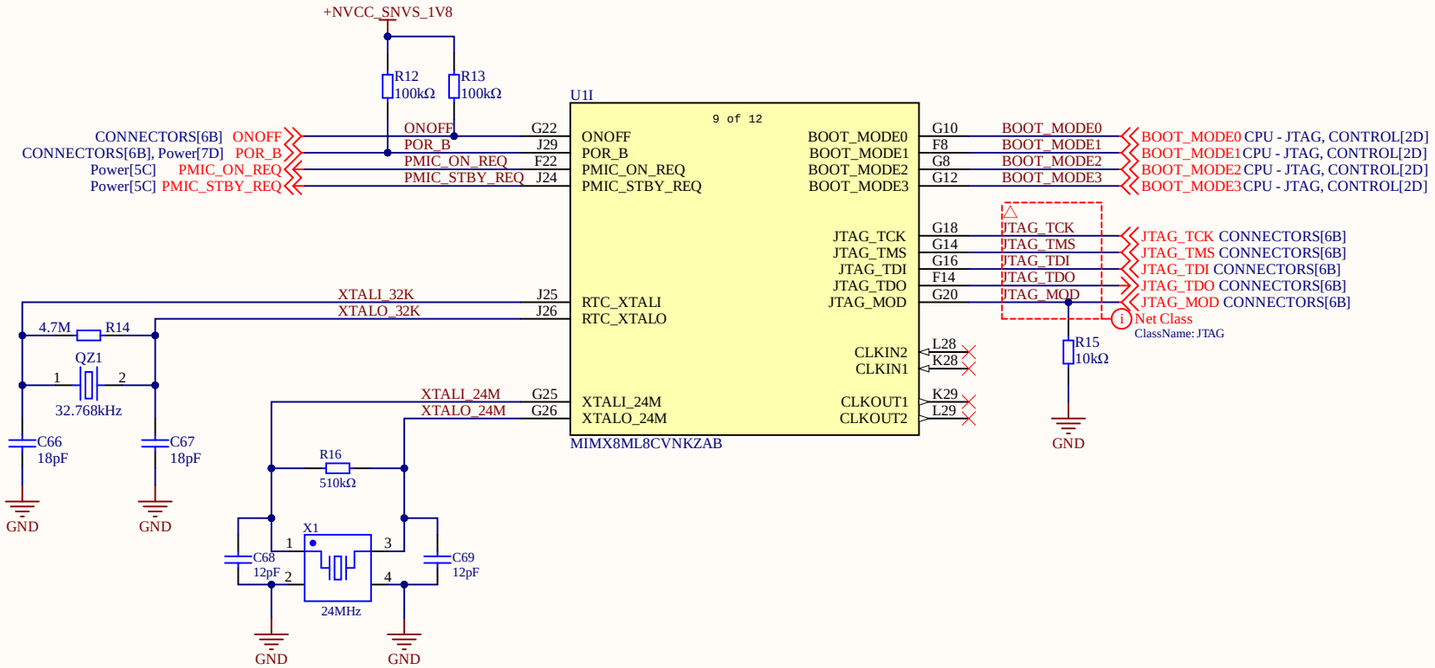
CPU - POWER

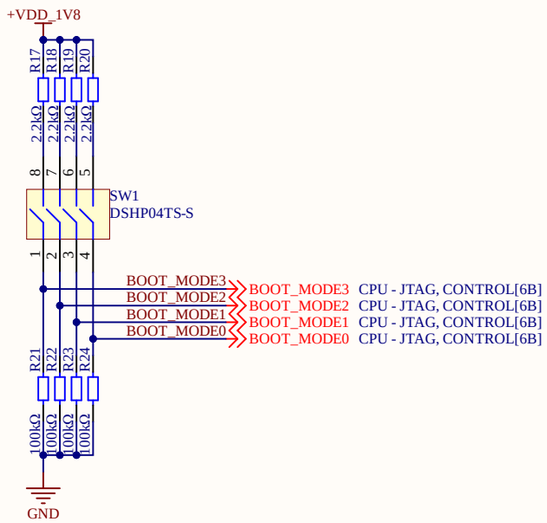


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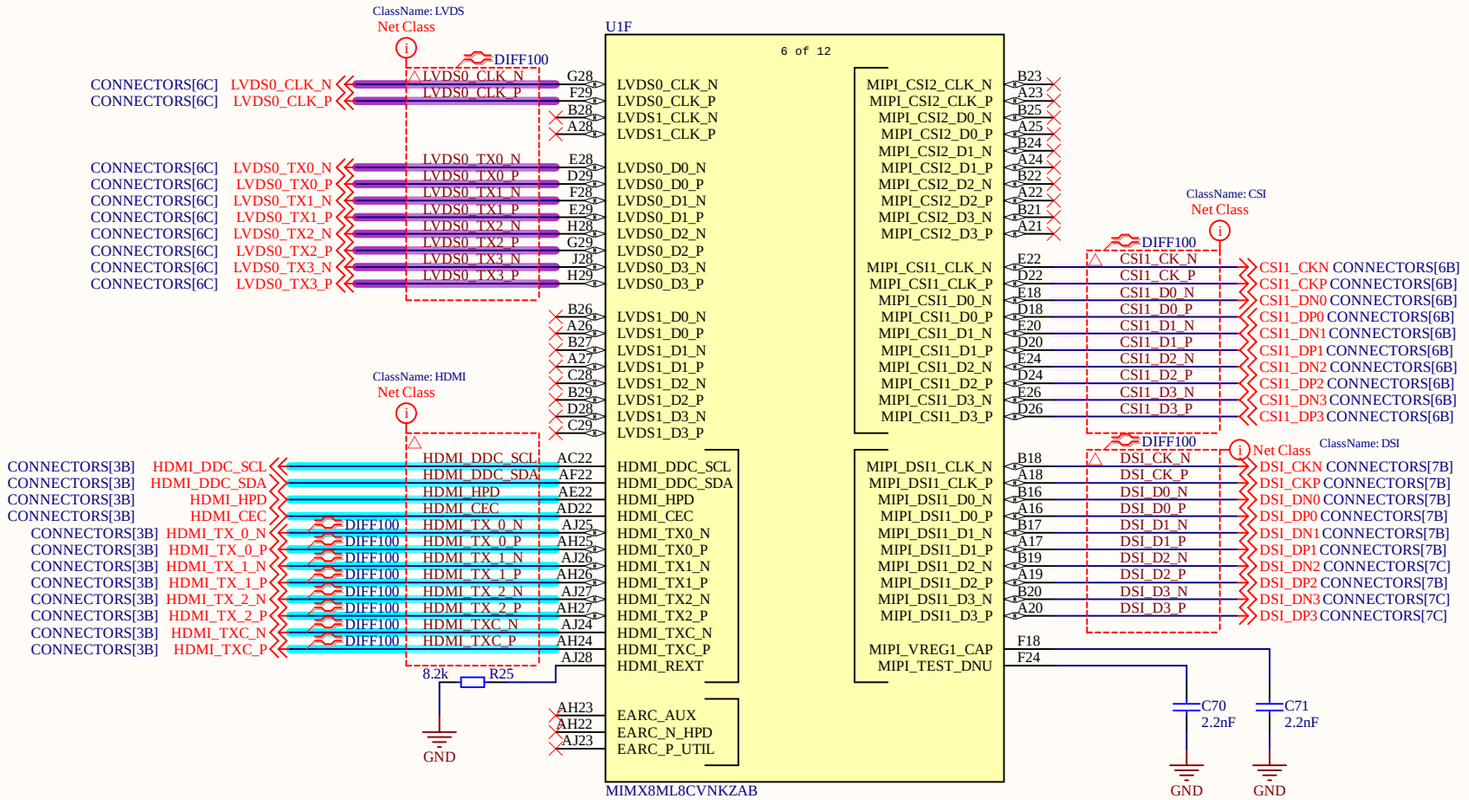
CPU - JTAG, CONTROL





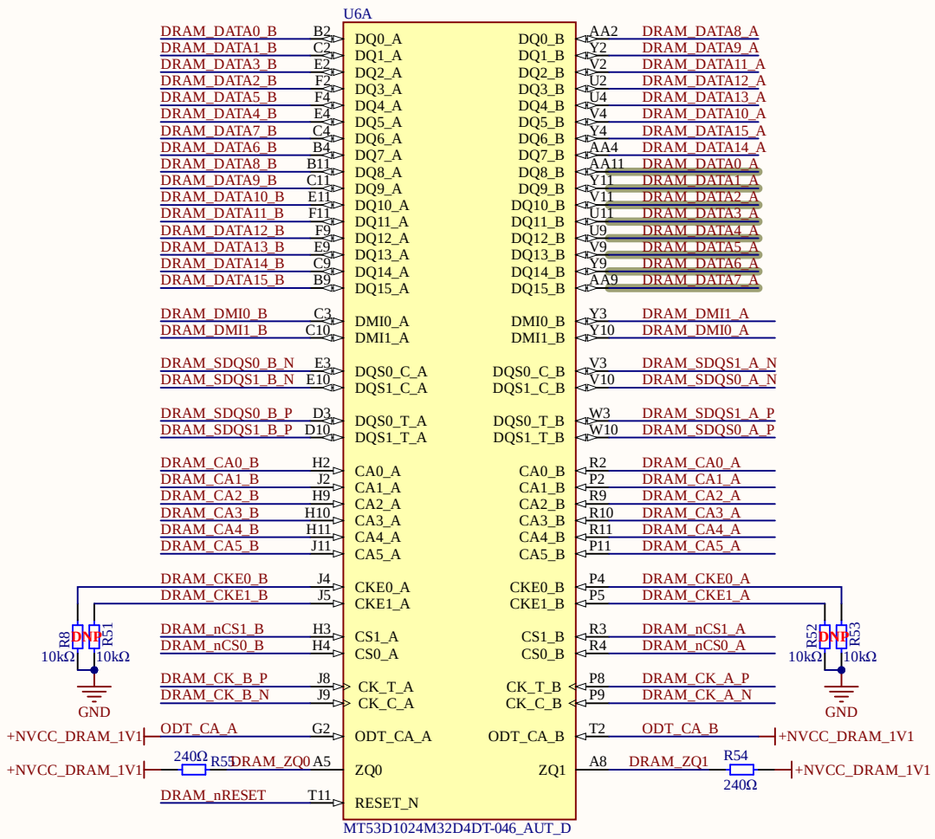
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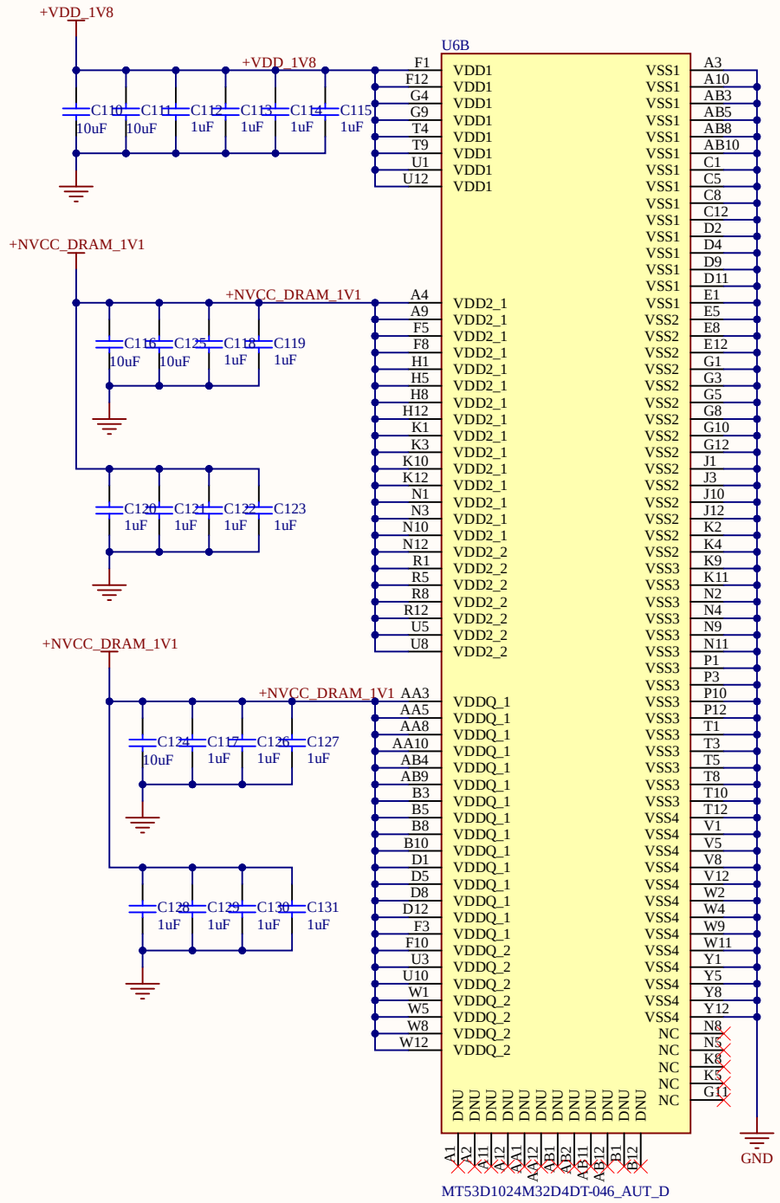
CPU - HDMI, LVDS, CSI, DSI



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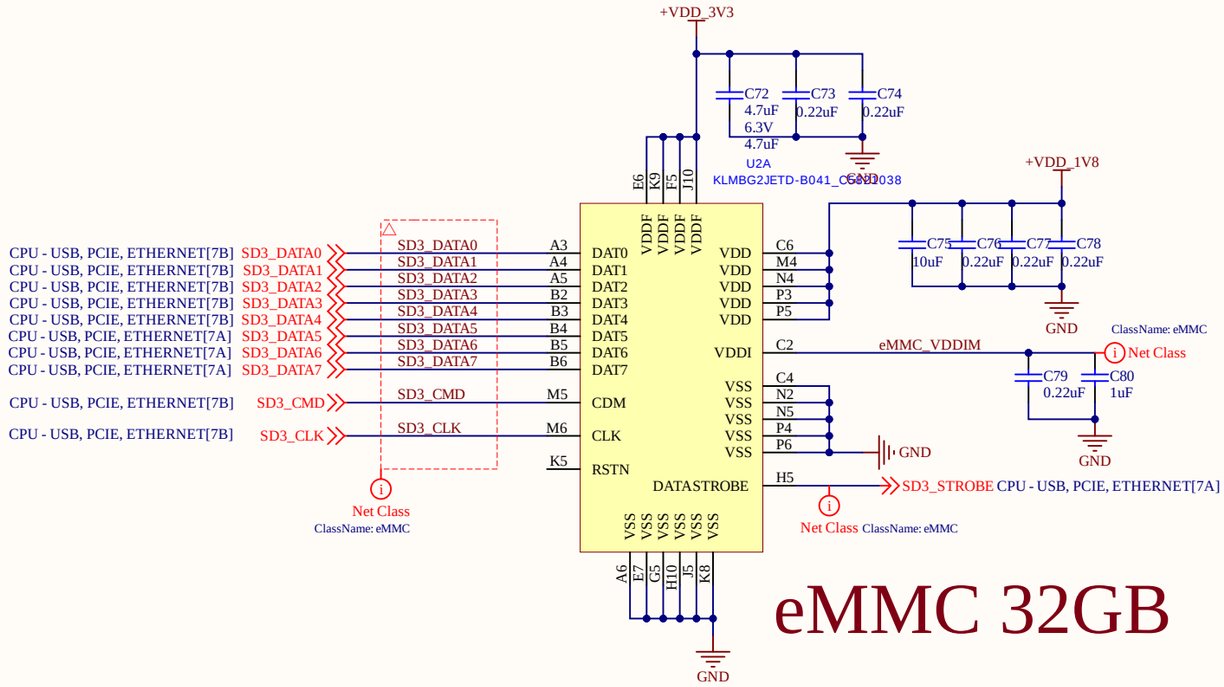
LPDDR4



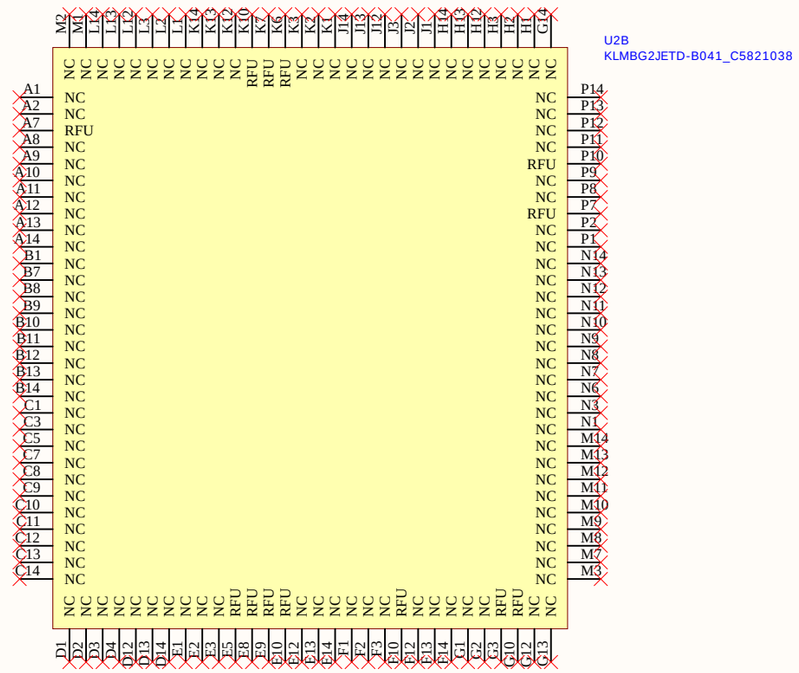


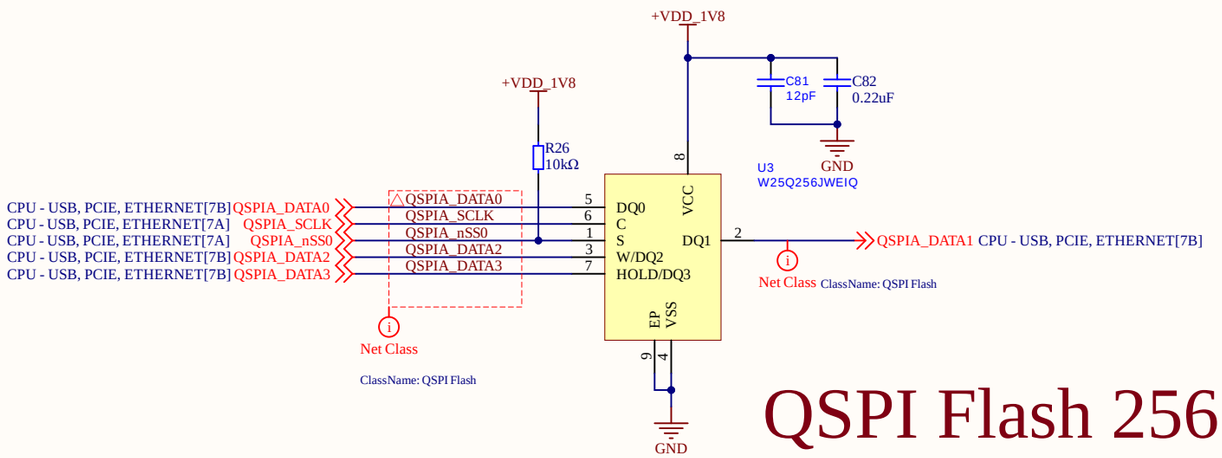
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eMMC, QSPI Flash



eMMC 32GB

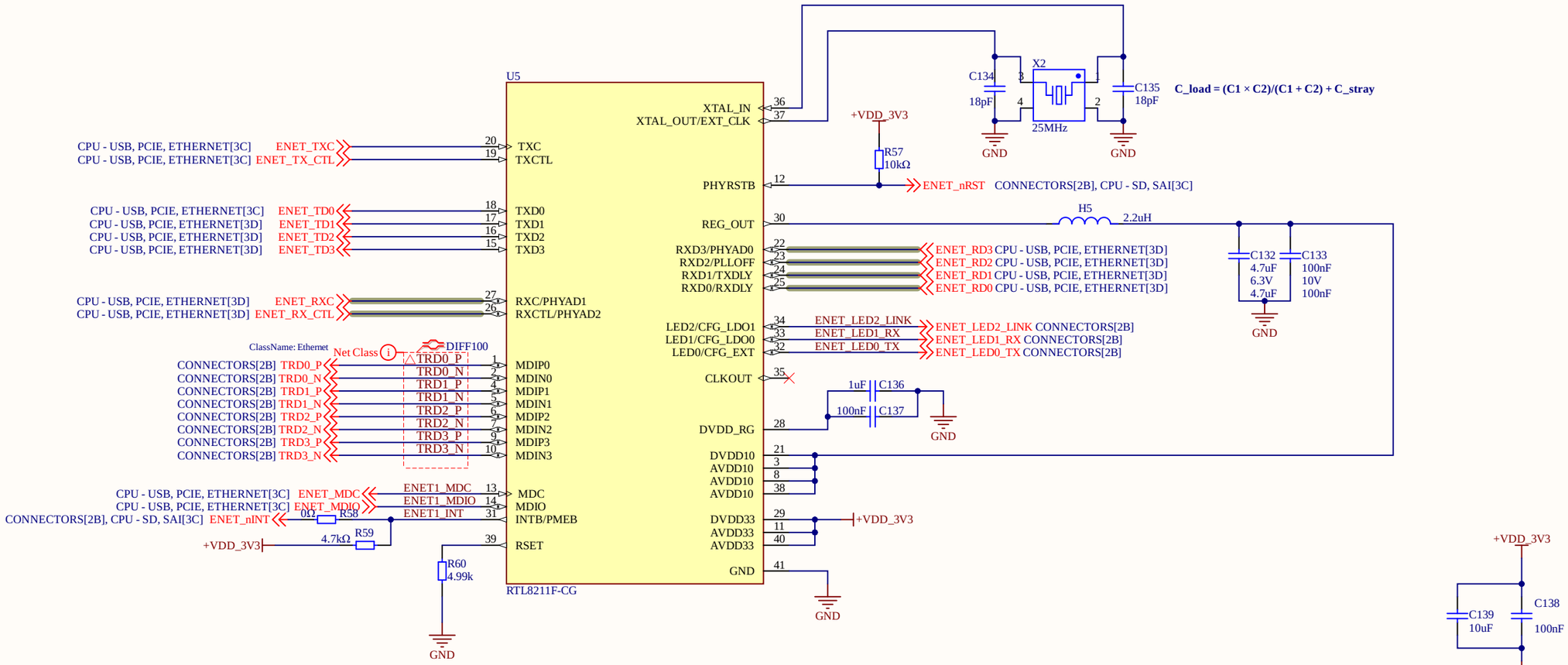




QSPI Flash 256 Mbit

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ETHERNET



9.1.1. PCB Layout

- The input 3.3V power trace connected to DVDD33 must be wider than 40mils
- The bulk de-coupling capacitors (Cin1 and Cin2) must be placed within 200mils (0.5cm) of DVDD33 to prevent input voltage overshoot
- The output power trace out of REG_OUT must be wider than 60mils
- Lx (2.2μH/4.7μH) must be kept within 200mils (0.5cm) of REG_OUT
- Cout1 and Cout2 must be kept within 200mils (0.5cm) of Lx to ensure stable output power and better power efficiency
- For switching regulator stability, the capacitor Cout1 and Cout2 must be a ceramic (X5R) capacitor. Cin1 and Cin2 are recommended to be ceramic capacitors
- Place Lx and Cin1 on the same layer as the RTL8211F(I). Do not use vias on DVDD33 and REG_OUT traces

Note: Violation of the above rules will damage the IC.

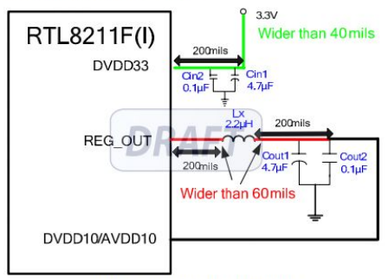
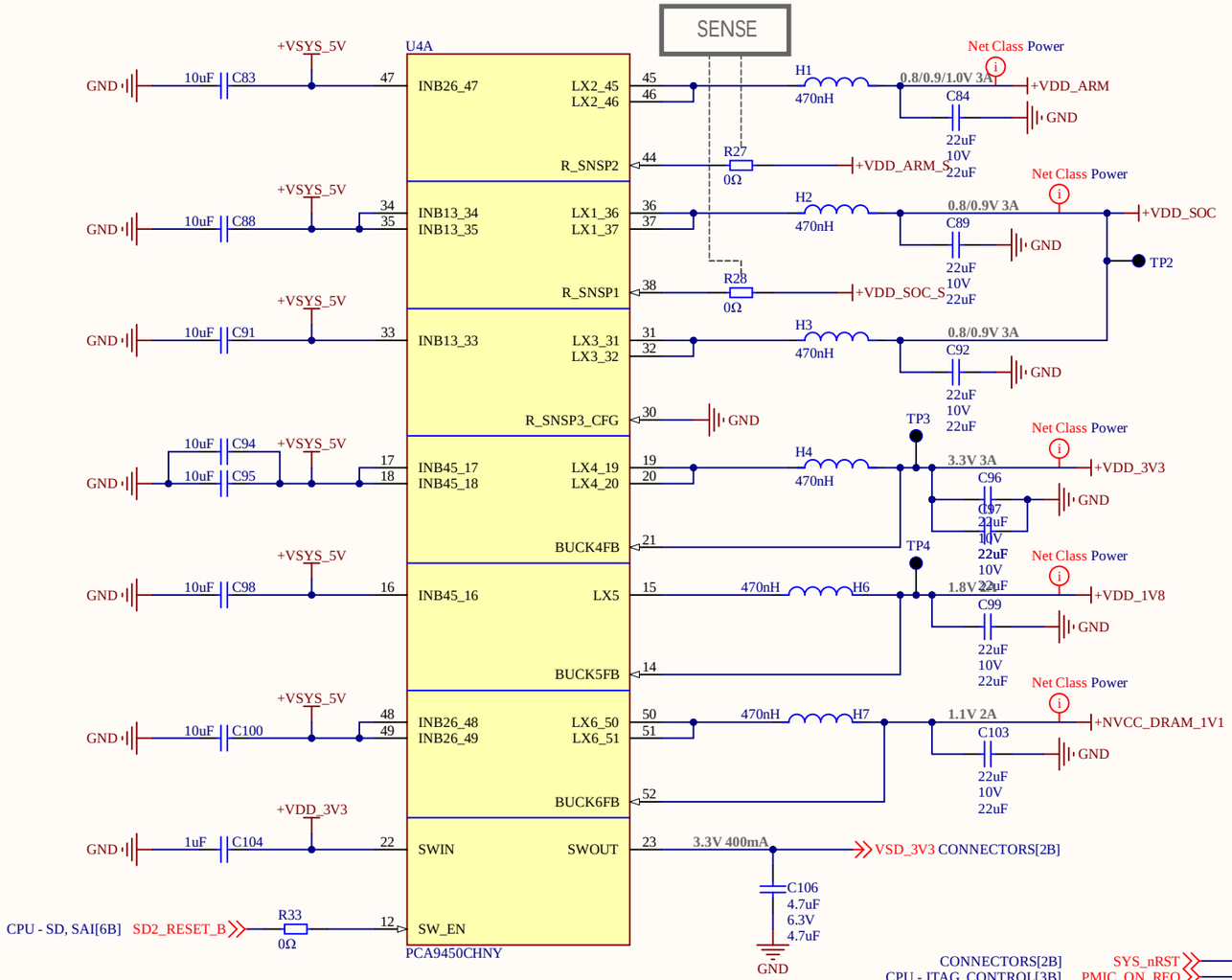
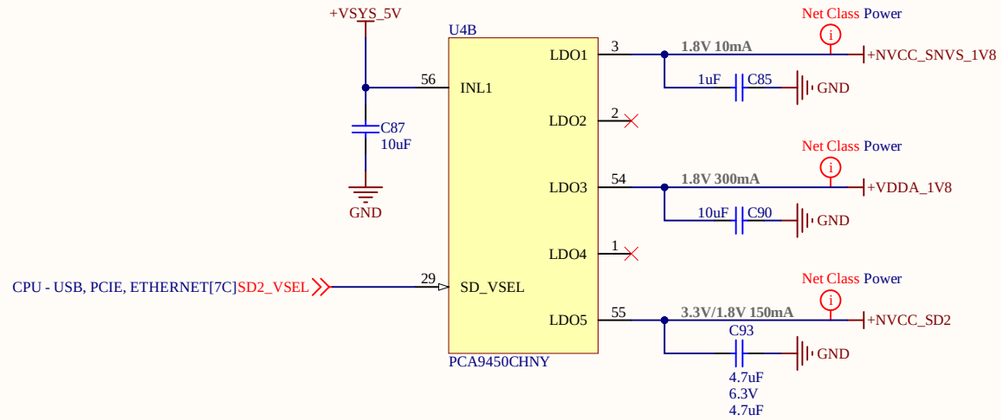


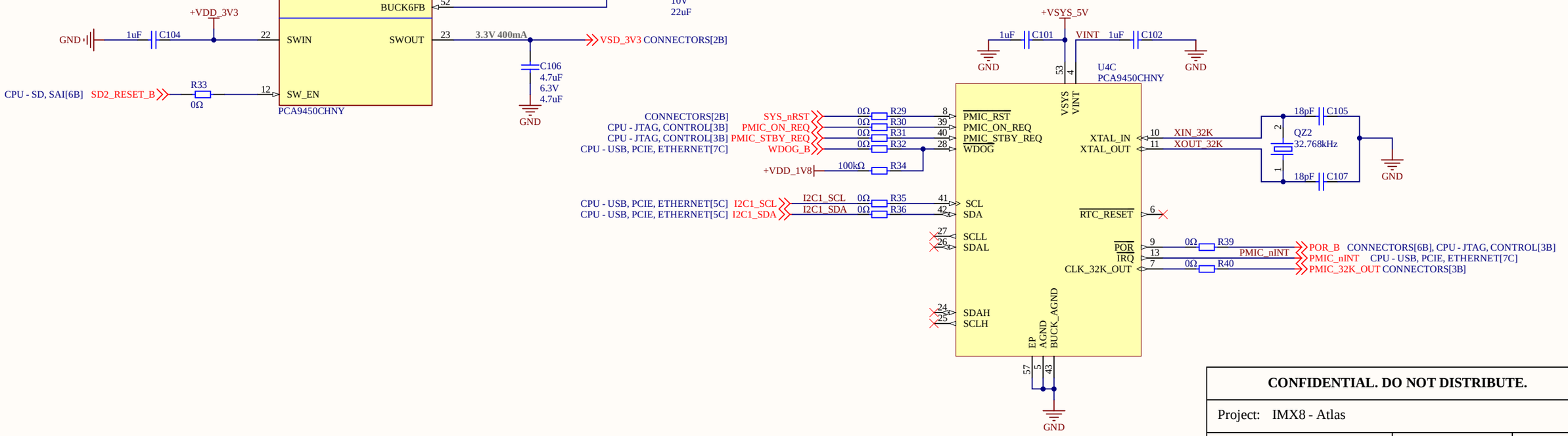
Figure 10. Switching Regulator

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POWER

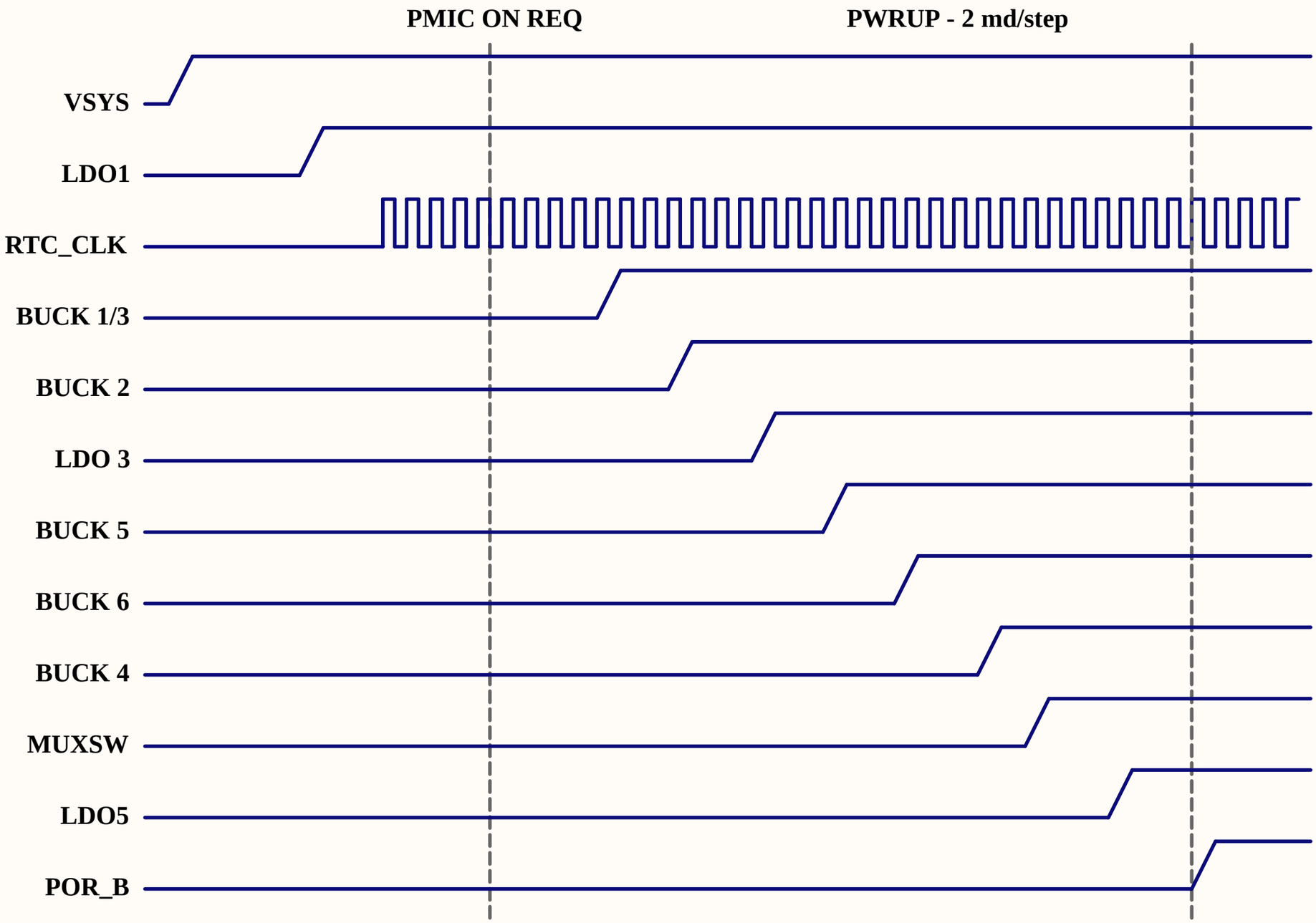






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POWER SEQUENCING



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REVISION HISTORY

DATE	COMMENT
28-AUG-2026	DRAFT FOR INITIAL REVIEW